An Effective Iterated Greedy Algorithm for Multi-Robot Task Allocation and Scheduling Problem

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***Abstract*—With the swift progress of intelligent and unmanned technologies, new opportunities and challenges have emerged for smart agriculture. The use of intelligent robots in various farming activities is becoming increasingly prevalent in agricultural production. This paper examines the issue of task allocation and scheduling for multiple agricultural robots working within the context of a smart farm. The multi-robot task allocation and scheduling problem (MRTASP) has been confirmed to be NP-hard problem. An effective iterated greedy (EIG) algorithm is presented to decrease the maximum completion time of the MRTASP in this study. In the EIG algorithm, to better exploit the specificity of the problem, the HPF2 algorithm is used to create an efficient initialization approach. A reconstruction method is proposed to improve the answer achieved at each iteration. The performance of EIG algorithm is tested on 720 instances. The EIG algorithm is useful for solving the MRTASP and its completion time is as short as possible, according to the results of the experiments.**

Keywords—multi-robot task allocation, scheduling, effective iterated greedy, maximum completion time

I. Introduction

Agriculture 4.0 signifies the onset of a transformative period in farming, distinguished by the embrace of intelligence, automation, and smart agriculture practices [1]. At the core of this revolution, smart agriculture has dramatically altered conventional farming methods through the integration of intelligent agricultural robots [2]. These robots are particularly adept at critical tasks such as harvesting and weeding, substantially enhancing the efficiency and precision of agricultural operations [3]. However, the complexity of effectively allocating and scheduling these robots remains a significant challenge. The challenge of complexity, along with considerations of operational costs and efficiency, has propelled our investigation into the task allocation and scheduling problem for multiple agricultural robots. The objective is to devise a solution that not only optimizes task distribution but also tackles the challenges of cost-effectiveness and operational efficiency.

The multi-robot task allocation and scheduling problem (MRTASP) is a complex, multifaceted challenge that involves assigning a set of tasks to a team of robots and determining the optimal sequence in which these tasks should be executed. This problem is inherently NP-hard [4], making it computationally intensive and difficult to solve optimally, especially as the number of robots and tasks increases. The complexity is further amplified by the need to consider constraints such as task dependencies, robot capabilities, and operational deadlines.

In the realm of smart farms, a fleet of n identical agricultural robots is initially stationed at a depot, awaiting directives from the control system to carry out assigned tasks. Each robot departs from the depot and returns upon the completion of its allocated tasks [5]. The incorporation of these robots into agricultural practices has resulted in a significant boost in productivity and a reduction in labor demands [6]. To further augment efficiency, it is imperative to explore effective methodologies for high-quality task allocation and scheduling that account for specific problem characteristics such as robot locations and task durations.

Over the past few decades, the MRTASP has garnered significant attention. Gerkey and Matarić [7] provided a foundational framework for this field by categorizing multi-robot task allocation based on the characteristics of robots, tasks, and allocation methods. Lee [8] presented a resource-based task allocation method for multi-robot systems, designed to boost efficiency and performance by optimizing the use of robotic resources. Dai et al. [9] developed a multi-objective optimization strategy using the discrete artificial bee colony algorithm. Guo et al. [4] introduced an efficient collaborative evolutionary algorithm for task allocation and scheduling of multiple agricultural robots in smart farms. Their approach, particularly for weeding tasks, improved system efficiency through an enhanced discrete artificial bee colony algorithm. Kang et al. [10] proposed a multi-objective teaching-learning optimization algorithm for task allocation among weeding robots in smart farming. The main goals were to reduce the longest task completion time and residual herbicide usage, thereby enhancing the overall efficiency and effectiveness of multi-robot operations. Dong et al. [11] developed an effective multi-objective evolutionary algorithm to address task assignments for multiple spraying robots. The aim was to optimize the total travel distance and maximum completion time for all robots operating in a greenhouse setting. In summary, the field of multi-robot task allocation and scheduling in agriculture has seen significant progress. Researchers have developed various methods to optimize robot tasks and schedules, focusing on efficiency, resource management, and operational costs. These advancements aim to make agricultural practices more efficient and sustainable through smarter use of robotic technology.

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This paper introduces an effective iterated greedy (EIG) algorithm designed to address the MRTASP in the context of smart agriculture. Our methodology augments the principles of greedy algorithms, renowned for their computational swiftness, and enhances them through an iterative process that refines solutions progressively. An EIG algorithm is developed to minimize the maximum completion time for the MRTASP in this study. The following is a list of the main contributions.

- The HPF2 approach is used to create an efficient initialization approach.
- To enhance the solution achieved at each iteration, a reconstruction method is proposed.
- Two local search approaches which improve local search capability are included in the improvement phase.

The remainder of this work is separated into the following sections. The description of the MRTASP and the mathematical model are presented in Section II. The EIG algorithm is presented in Section III. The experimental results and comparisons are presented in Section IV. Section V contains the conclusion as well as some recommendations for future research.

II. Formulation of MRTASP

A. Description of the Problem

In this research, we tackle the challenge of assigning and sequencing weeding tasks in a smart agricultural setting, where a small team of weeding robots (n) is tasked with managing a large number of weeding tasks (m) across various regions, under the condition that n is significantly less than m. We preclude the assignment of a single task to multiple robots, as such distributions are deemed impractical. The primary goal is to determine an optimal set of task allocations and sequences for each robot that maximizes the overall completion time most efficiently. Furthermore, to simplify our model without compromising its general applicability, we make the following assumptions:

(a) All robots function without failure or breakdown.

(b) The robots are identical in capabilities.

(c) Each robot initiates and concludes its operations at a central depot.

(d) Robots travel at a consistent speed, the value of which is predefined and known.

B. Mathematical Model of the MRTASP

Notation:

v	Speed of the robot.
k	Index of robots.
i, j, l	Task index.
x_i, y_i	Coordinate values of the depot and task point in the set J.
t_i	Weeding time required for task point i, and $t_i = 0$.
$D_{i,j}$	The distance from task J_i to task J_j.
π_k	The sequence of tasks performed by robot.
R	$R = \{R_1, R_2, \dots, R_n\}$. The collection comprises n robotic units.
J	$J = \{J_0, J_1, J_2, \dots, J_m\}$. A set of depot and task points consisting of m locations awaiting weeding operations. J_0 is the depot.
C_k	The time required for the robot R_k to return to the depot after performing all its assigned tasks.
C_{max}	The time for completion of all tasks.

Decision variables:

$Y_{i,k}$	A binary variable that indicates whether or not the robot R_k executes the task J_i. 1 means execute and 0 means do not execute.
$X_{i,j,k}$	Binary variable which denotes whether or not the robot R_k executes task J_j after task J_i. 1 means execute and 0 means do not execute.
$S_{i,j,k}$	If robot R_k travels from task J_i to task J_j, its value is the number of tasks that robot R_k has performed before task J_i, otherwise it is 0.

Objective and constraints:

$$MinC_{max} \tag{1}$$

$$\sum_{k=1}^{n} Y_{i,k} = 1, \forall i \backslash \{0\} \tag{2}$$

$$Y_{0,k} = 1, k = 1,2,\dots,n \tag{3}$$

$$\sum_{j=0, j\neq i}^{m} X_{i,j,k} = Y_{i,k}, i = 0,1,\dots,m, \forall k \tag{4}$$

$$\sum_{i=0, i\neq j}^{m} X_{i,j,k} = Y_{j,k}, j = 0,1,\dots,m, \forall k \tag{5}$$

$$S_{i,j,k} \leq X_{i,j,k}\left(\sum_{l=0}^{m} Y_{l,k} - 1\right), \forall k, i \neq j \tag{6}$$

$$\sum_{j=0}^{m} S_{i,j,k} - \sum_{j=0}^{m} S_{j,i,k} = 1, \forall i \backslash \{0\}, \forall k \tag{7}$$

$$C_k = \sum_{i=0}^{m} \sum_{j=0, j\neq i}^{m} \left(X_{i,j,k} \cdot \frac{fabs(x_j - x_i) + fabs(y_j - y_i)}{v} \right) + \sum_{i=0}^{m} (Y_{i,k} \cdot t_i), \forall k \tag{8}$$

Equation (1) defines our objective function, which is to minimize the completion time of the last robot to finish its tasks. Constraint (2) ensures that each task is exclusively assigned to one robot. Constraint (3) confirms that every robot begins from the depot. Constraints (4) and (5) maintain the balance of task sequencing. Constraint (4) guarantees that if robot R_k performs task J_i, it must then proceed to a unique subsequent task (not J_i), establishing a clear succession. Similarly, Constraint (5) ensures that if robot R_k performs task J_j, there is a unique preceding task (other than J_j itself) that R_k must complete before undertaking J_j. Constraints (6) and (7) are crafted to prevent the emergence of subtours, preserving the coherence of the task allocation. Lastly, constraint (8) calculates the completion time for robot R_k, encapsulating the sequence and duration of its assigned tasks.

III. An Iterated Greedy Algorithm

The iterated greedy algorithm was devised by Ruiz and Stutzle [12] to solve the problem of permutation flowshop scheduling. The EIG algorithm is intended to make the C_{max} as short as possible for the MRTASP in this study. The framework of EIG algorithm is depicted in Algorithm 1. The EIG algorithm begins by performing an initialization step. Second, to improve the initial solution. Swapping and inserting neighborhood structures (LS1 and LS2, respectively) are used in a variable neighborhood search (VNS). The destruction, reconstruction, local search, and acceptance phases of the iterative process establish a new solution.

Algorithm 1: Proposed EIG

π = Initialization Phase
$\pi^b = \pi$
While condition for termination not met **do**
$\pi^* = \pi$
π_E = Destruction(π^*, π_R)
π^* = Reconstruction(π_R, π_E)
π^* = Successively apply LS1, LS2 on π^*
π = Solution update(π^*, π, π^b)
End while
Return π^b

A. Initialization Phase

The population is initialized with the HPF2 strategy [13]. The HPF2 regulation is split into two parts. To begin, choosing a task to be the first in the order. Second, constructing the remainder of the sequence to reduce robot downtime and optimize the significance of each task to extend the C_{max}. As a result, HPF2 has the following definition:

Step.1: Choose the task with the lowest $R(i)$, as determined in Eq. (9), and assign it first in the sequence π_k. Set $k = 1$.

Step.2: While $k < n$, the task with the minimum $ind(i,k)$, according to Eq. (10), was picked and assigned to all potential spots in the current sequence of each robot. Choose the task that leads to the shortest C_{max} partial sequence, $k = k + 1$.

$$R(i) = \lambda \cdot 2 \cdot \sum_{j=1}^{m} (m-j) \cdot D_{i,j}/(m-1) + (1-\lambda) \cdot \sum_{j=1}^{m} D_{i,j} \tag{9}$$

$$ind(i,k) = \mu \cdot \sum_{j=1}^{m} \left(D_{j,k}(\sigma * i) - D_{j,k-1}(\sigma) - D_{i,j}\right) + (1-\mu) \cdot \left(D_{m,k}(\sigma * i) - D_{m,k-1}(\sigma)\right) \tag{10}$$

B. Improvement Phase

The improvement phase includes two local search algorithms, LS1 and LS2, which improve local search capacity. The following is a description of the LS1 technique. By examining the present sequence, LS1 attempts to optimize task sequences at each robot. If the better solution outperforms the current one, it becomes the new solution, and the cycle repeats until all tasks are accomplished. The task is allocated at random to prevent exploring the neighborhoods in the same order every time.

LS2 has agreed to relocate tasks from the crucial robot to other robots. Neighbors are created for each task by deleting the task from its present position in the order and placing it in all other feasible locations. Tasks are chosen at random, much like in LS1. If the solution proposed by the most wonderful neighbor is better than the existing one, it replaces the current one, and the process is repeated until all tasks are taken into account.

C. Destruction Phase

The destruction approach involves interfering with the current solution to create a new search location. During this stage, a certain number of tasks are stripped away from the current solution π. No duplication and a group of tasks is referred to as π_R. The solution π after the procedure for removing the object has been identified as π_E. From the sequence of candidate solutions, a variety of D tasks are generated.

D. Reconstruction Phase

Using a greedy mechanism, each task from π_E is reinserted into π_R during this stage. In all robots, the task is checked in all conceivable positions. Finally, after inserting the task, the chosen location is the one that provides the sequence with the lowest C_{max}. When the task is reinserted into task location k for the first time, an operator is required to refine the solution. By comparing all potential positions, choose a task at random before or after the task ($k + 1$ or $k - 1$) with equal probability. The selected work is moved to the position that reduces the maximum completion time. When all tasks of π_E are reinserted, a thorough candidate solution is obtained. Algorithm 2 depicts the reconstructing strategy.

Algorithm 2: Reconstruction ($\pi_R = \{\pi_1, \cdots, \pi_F\}, \pi_E$)

For i = 1 to $|\pi_E|$
  $j = \pi_E[i]$
  For $k = 2$ to m **Do**
  test task j in every conceivable posture of π_f
  End For
  $k_{min} = \arg(min(C_{max}))$
  insert j in k_{min} of π_{min}
  extract j^* from k_{min}+1 or k_{min}-1
  test j^* in every conceivable posture of π_R
  insert j^* in the smallest position C_{max}
End For
Return π_R

E. Acceptance Criterion

Due to insufficient diversification, the EIG metaheuristic implementation of this acceptance criterion may result in a static state of search. As a result, we propose simulated remelting with a uniform temperature as a simple acceptability criterion. Three steps of a proposed solution were generated. The constant temperature is determined as follows in Eq. (11).

$$Temperature = T \cdot \frac{\sum_{i=1}^{m}\sum_{j=1}^{n} D_{i,j}}{n \cdot m \cdot 10} \quad (11)$$

Where T is the parameter that has to be modified.

IV. Experiments and Comparisons

The experimental benchmark set has different combinations of $n \times m$: $\{(20,50,100) \times (5,10,20), 200 \times (10,20), 500 \times 20\}$. Each scale has 10 different instances. All of the experiments in this paper make use of the greatest amount of CPU time $20 \times n \times m$ as termination guidelines. Each instance is performed five times with various parameter combinations each time on a computer with a 2.30GHz Intel Core i5-6300 processor and 4.0GB of RAM. We set $d = 4$, $L = 30$ for the following algorithm is used to evaluate EIG algorithm.

Three algorithms are compared to the proposed EIG algorithm: modified iterated greedy algorithm (MIG) [14], bounded-search iterated greedy algorithm (BSIG) [15], and the two stage iterated greedy method (IG2S) [16]. The test problem consists of 720 cases that are run five times each by each of the three algorithms separately. The outcomes are measured using the average relative percentage deviation (ARPD) index [17], with the computation technique provided in Eq. (12). Other methods are outperformed by the algorithm with the lowest ARPD.

$$ARPD = \frac{1}{R} \cdot \sum_{i=1}^{R} \frac{C_i - C_{opt}}{C_{opt}} \cdot 100\,\% \quad (12)$$

Where the solution provided by the algorithm used during the ith experiment of the given case is denoted by C_i. The quantity of runs is denoted by R. The smallest C_{max} discovered by algorithms is C_{opt}. The APRD are expressed as percentage values and the % is omitted from the charts.

TABLE I. ARPD of Algorithms MIG, BSIG, IG2S, and EIG

Instance		MIG	BSIG	IG2S	EIG
n	20	0.045	0.032	0.026	**0.015**
	50	0.624	0.612	0.414	**0.055**
	100	0.897	0.581	0.317	**0.121**
	200	0.643	0.292	0.144	**0.066**
	500	0.162	0.130	0.101	**0.045**
m	5	0.620	0.465	0.317	**0.074**
	10	0.927	0.576	0.334	**0.080**
	20	0.955	0.640	0.370	**0.168**
Average		0.522	0.347	0.208	**0.105**

All algorithms generated ARPD values are grouped by robot and task as demonstrated in TABLE I. In terms of overall performance, the findings indicate that the EIG algorithm proposed in this study outperforms other algorithms. The lower the ARPD value, the better the outcome. In addition, Fig. 1 gives the 95 % confidence level intervals for the least significant difference (LSD) generated by the multifactor analysis of variance (ANOVA) for the overall mean plot. The confidence intervals of the EIG algorithm do not coincide with those of the MIG, BSIG, and IG2S algorithms. Fig. 1 shows that the EIG algorithm outperforms the others statistically.

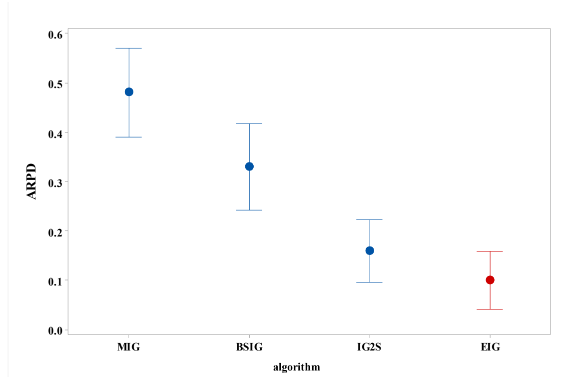

Fig. 1 Mean plot with 95% confidence intervals for algorithm MIG, BSIG, IG2S, and EIG

A mean plot with 95 percent confidence level intervals for algorithms with a large number of instances $(n \times m)$ is shown in Fig. 2. As ever observed, the ARPD value of MIG and EIG are equivalent when the number of tasks is 20. Minimum ARPD worth for EIG algorithm among the remaining tasks. In general, EIG algorithm outperforms other algorithms. The number of robots also has an effect, with the ARPD of all algorithms tending to drop as m increases.

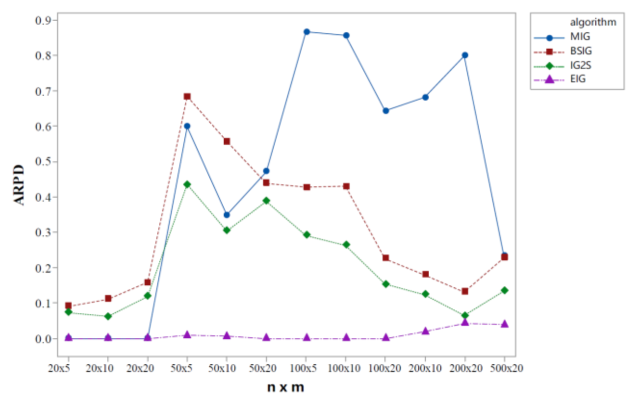

Fig. 2 ARPD values of algorithms with $n \times m$

From the TABLE II. and Fig. 3, the Friedman test [18] was included to see whether the means of the four error algorithms differed significantly. The EIG algorithm has the smallest average ranking of the four algorithms in terms of statistical findings with $\alpha = 0.05$ and $\alpha = 0.10$ as seen in the boxplots. The EIG algorithm is statistically superior than the other algorithms and performs well in solving the MRTASP, as shown by the aforementioned experimental results.

TABLE II. Test of Friedman

Algorithms	Mean Rank
MIG	3.830
BSIG	2.830
IG2S	1.830
EIG	1.500
Crit.Diff.$\alpha = 0.05$	1.6773
Crit.Diff.$\alpha = 0.10$	1.6528

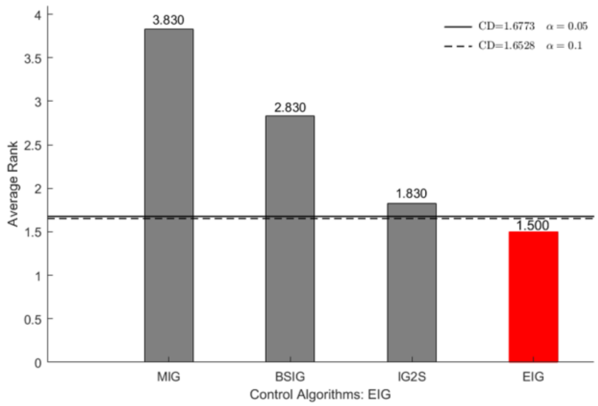

Fig. 3 Friedman test rankings

V. Conclusion and Future Research

An efficient iterated greedy algorithm is presented for the MRTASP with the shortest maximum completion time requirement among all robots in this work. The suggested EIG algorithm uses the HPF2 initialization approach to apply to a new task distribution rule. Then, destruction phase, an enhanced reconstruction phase and improvement phase designed for the problem. A simple acceptance condition for annealing-like simulation at a temperature to ensure is also added to aid the algorithm in escaping from the local best solution. The current EIG algorithm is assessed using 720 benchmark instances and compared to MIG, BSIG, and IG2S algorithms. The experimental results show that the proposed EIG algorithm performs better other algorithms. Out of 720 occurrences, 552 best solutions were updated.

In future research, the MRTASP will be integrated with other optimization goals. Existing scheduling studies have been carried out under ideal conditions; in real problems, the constraints are much more complex. It will be more relevant to study scheduling with different constraints and characteristics for different problems. Consideration of integration with actual production is also an intriguing field of inquiry.

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